called cytokine which activates B cells. Killer cells move around the body and destroy cells which are damaged or infected **(Fig. 7.14)**.

Apart from these cells neutrophils and monocytes destroy foreign cells by phagocytosis. Monocytes when they mature into large cells, they are called macrophages which perform phagocytosis on any foreign organism.

Dendritic cells are called so because its covered with long, thin membrane extensions that resemble dendrites of nerve cells. These cells present the antigen to T-helper cells. Four types of dendritic cells are known. They are langerhans, interstitial cells, myeloid and lymphoid cells

7.3.5 Antigens

The term **antigen** (Ag) is used in two senses, the first to describe a molecule which generates an immune response and the second, a molecule which reacts with antibodies. In general antigens are large, complex molecular substances that can induce a detectable immune response. Thus an antigen is a substance that is specific to an antibody or a T-cell receptor and is often used as a synonym for immunogen.



The histocompatibility antigens are cell surface antigens that induce an immune response leading to rejection of allografts.

An **immunogen** is a substance capable of initiating an immune response. **Haptens** are substance that are non-immunogenic but can react with the products of a specific immune response. Substances that can **enhance the immune response** to an antigen are called **adjuvants**. **Epitope** is an antigenic determinant and is the active part of an antigen. A **paratope** is the antigen – binding site and is a part of an antibody which recognizes and binds to an antigen.



Antigenicity is the property of a substance (antigen) that allows it to react with the products of the specific immune response.

Types of antigens

On the basis of origin, antigens are classified into **exogenous antigens** and **endogenous antigens**.

The antigens which enter the host from the outside in the form of microorganisms, pollens, drugs, or pollutants are called **exogenous antigens**. The antigens which are formed within the individual are **endogenous antigens**. The best examples are blood group antigens.

7.3.6 Antibodies

Antibodies are immunoglobulin (Ig) protein molecules synthesized on exposure to antigen that can combine specifically with the antigen. Whenever pathogens enter our body, the B-lymphocytes produce an army of proteins called antibodies to fight with them. Thus, they are secreted in response to an antigen (Ag) by the effect of B cells called plasma cells. The antibodies are classified into five major categories, based on their physiological and biochemical properties. They are **IgG** (gamma), **IgM** (mu), **IgA** (alpha), **IgD** (delta) and **IgE** (epsilon).

In the 1950s, experiments by **Porter and Edelman** revealed the basic structure of the immunoglobulin. An antibody molecule is **Y** shaped structure that comprises of four polypeptide chains, two identical light chains **(L)** of molecular weight 25,000 Da (approximately 214 amino acids) and two identical heavy chains **(H)** of molecular weight 50,000 Da (approximately 450 amino acids). The polypeptide chains are linked together by di-sulphide (S-S) bonds. One light chain is attached to each heavy chain and two heavy chains are attached to each

XII Std Biology-Zoology Chapter-7 EM.indd 123 23/12/2022 12:39:58